Six-Functor Formalisms over Orbit Categories

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Abstract

[Sch26b] and [Sch26a]

Contents

1 Equivariant Homotopy Theory and Orbit Category

Let G denote a compact Lie group.

Definition 1.1. The orbit category $\mathcal{O}_{\mathrm{G}}$ associated to G is defined by

Objects $\{G/H \mid H \text{ is a closed subgroup of } G\}$

Morphsims Continuous G-equivariant maps.

Let $\mathrm{Hom}_{\mathcal{O}_{\mathrm{G}}}(G/H, G/K)$ equipped with compact-open topology, then we have the homeomorphsim

$$\mathrm{Hom}_{\mathcal{O}_{\mathrm{G}}}(G/H, G/K) \simeq (G/K)^H.$$

Definition 1.2. An $\mathcal{O}_{\mathrm{G}}$-space is a contravariant functor from $\mathcal{O}_{\mathrm{G}}$ to Top.

Definition 1.3. A G-space is a covariant functor from classifying category BG to Top.

Theorem 1.4 (Elmendorf's Theorem)**.**

References

[Sch26a] Peter Scholze. Geometry and higher category theory, 2026. Lecture notes for course WS 2025/26.

[Sch26b] Peter Scholze. Six-functor formalisms, 2026.